

Intro to Accessibility and Aging

IN4MATX 232

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What Is **Accessibility**?

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Accessibility is broadly defined as the degree to which ***a product, device, service, or environment is available to as many people as possible***. In the context of digital systems, the World Intellectual Property Organization (WIPO) and the International Standards Organization (ISO) define accessibility in terms of the prevention of activity limitations and the facilitation of participation for people with disabilities. Historically, accessibility was viewed as a ***technical compliance issue rooted in the medical model of disability***.

What Is **Assistive Technology**?

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Assistive Technology has traditionally been defined through a legislative and medical lens. The Technology-Related Assistance for Individuals with Disabilities Act of 1988 (the "Tech Act") provided the first official definition in the United States, describing AT as ***any item, piece of equipment, or product system used to increase, maintain, or improve the functional capabilities of individuals with disabilities.***

The reauthorization of the Assistive Technology Act in 2004 marked a key shift, ***placing individuals with disabilities at the forefront of service planning and expanding the scope to include "electronic and information technology"***. Today, the WHO utilizes AT as an "umbrella term" encompassing both products and the related systems and services that help maintain functioning in cognition, communication, hearing, mobility, self-care, and vision.

(Special) Education

Geriatric Medicine

Disability Studies

Biomedical Engineering

Inclusive Design

**Audiology & Speech-
Language Pathology**

Accessibility & Aging

Human-Computer Interaction

Neuropsychology

Urban Planning & Architecture

Clinical Psychology

Public Policy

Assistive Technology

The History of the Accessibility and Aging Subcommittee at CHI

Year	Subcommittee Name	Relationship to Aging/Accessibility
2009-2013	Usability, Accessibility and User Experience	Integrated with Usability; Aging in "Specific Applications".
2014-2016	Specific Application Areas	Accessibility and Aging both moved to "Specific Applications".
2017-2018	Health, Accessibility and Aging	Combined Health and Accessibility into a dedicated track.
2019-Present	Accessibility and Aging	Standalone subcommittee for Accessibility and Aging.

CHI 2026

This subcommittee is suitable for contributions related to the design or study of technology for people with disabilities and/or older adults. Accessibility papers...technology designed for or used by people with disabilities ***including sensory, motor, mobility, psychosocial or cognitive, intellectual or learning disabilities, or people who identify as neurodivergent.*** Aging papers are ...used by ***people in the later stages of life***....We welcome contributions across a range of topics aimed at benefiting relevant stakeholder groups and not solely limited to concerns of making technology accessible.

CHI 2026 (continued)

Note that if your paper primarily concerns health outcomes or interactions with health data or with healthcare providers, then the Health_subcommittee is probably a better fit, whereas papers reflecting on how technologies are used or designed for specific needs are a better fit for this subcommittee....***This subcommittee welcomes all contributions related to accessibility and aging, including empirical, theoretical, conceptual, methodological, design, and systems contributions.***

In-Scope vs Out-of-Scope

Health Outcomes: At CHI, if a paper primarily concerns health outcomes or interactions with health data/providers, it is generally considered a better fit for the Health subcommittee.

Beyond Technology: The CHI subcommittee welcomes papers that are "not solely limited to concerns of making technology accessible," such as studies on how older adults learn programming or choose not to participate in social interventions.

CHI 2026 (continued)

Your paper should also use **inclusive language**. Avoid characterizing an entire population using phrases that represent outlier positions for individuals, like “suffering from” (negative) or “inspirational” (positive). When comparing across characteristics or experiences, avoid referring to some people as “normal” implying others are not, or broadly characterizing one group’s experiences as categorically better or worse than others. We also recommend avoiding terms like “vulnerable,” “special needs,” and “X challenged.” We understand there may be exceptions (e.g., medical vision classification systems define a visual acuity as “normal”), but such usage should be footnoted for clarity, and these measures should be relevant to the described work.

Examples:

- Discovering Accessible Data Visualisations for People with ADHD
- Analyzing Accessibility Reviews Associated with Visual Disabilities or Eye Conditions
- Envisioning the (In)Visability of Discreet and Wearable AAC Devices
- Technology Adoption and Learning Preferences for Older Adults: Evolving Perceptions, Ongoing Challenges, and Emerging Design Opportunities
- Screen Recognition: Creating Accessibility Metadata for Mobile Applications from Pixels
- Understanding Older Adults' Participation in Design Workshops
- The Promise of Empathy: Design, Disability, and Knowing the “Other”
- “If It’s Important It Will Be A Headline”: Cybersecurity Information Seeking in Older Adults
- Older People Inventing their Personal Internet of Things with the IoT Un-Kit Experience
- Addressing Age-Related Bias in Sentiment Analysis

ASSETS

The International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) remains the undisputed premier forum for ***research concerning the design, evaluation, use, and education related to computing for people with disabilities and older adults***. Since its inception in 1994, ASSETS has fostered an environment where high-quality original submissions can explore both mainstream and specialized technologies.

ASSETS

Submission Track	Primary Objective
Technical Papers (Long)	Present mature, significant contributions to design, systems, or scientific understanding.
Technical Papers (Short)	Focused research contributions, such as a novel interaction technique or a pilot study.
Experience Reports	Reflection on the application of accessibility in education, industry, or policy settings.
Posters and Demos	Interactive showcasing of prototypes or preliminary research findings.
Doctoral Consortium	Mentorship for PhD students to refine their research focus and methodology.
Student Research Comp.	Competition for students to present original research in a professional setting.

CSCW

The ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) approaches accessibility through the lens of social interaction and collaboration. For a paper to be in scope for CSCW, it must focus on "social aspects of technology mediation" and be properly contextualized in the CSCW literature, ***referencing concepts such as collective intelligence, crowdsourcing, or remote collaboration***. A critical distinction at CSCW is the definition of "out of scope" work. Research contributions that are ***primarily of relevance to individual users are considered out of scope***. This stands in contrast to ASSETS, which frequently accepts work focusing on individual assistive technologies.

- **Social and crowd computing.** Studies, theories, designs, mechanisms, systems, and/or infrastructures addressing social media, social networking, wikis, blogs, online gaming, crowdsourcing, collective intelligence, virtual worlds, or collaborative information behaviors.
- **CSCW and social computing system development.** Hardware, architectures, infrastructures, interaction design, technical foundations, algorithms, and/or toolkits that are explored and discussed within the context of building new social and collaborative systems and experiences.
- **Methodologies and tools.** Novel human-centered methods, or combinations of approaches and tools used in building collaborative systems or studying their use.
- **Critical, historical, ethnographic analyses.** Studies of technologically enabled social, cooperative, and collaborative practices within and beyond work settings illuminating their historical, social, and material specificity, and/or exploring their political or ethical dimensions.
- **Empirical investigations.** Findings, guidelines, and/or studies of social practices, communication, cooperation, collaboration, or use, as related to CSCW and social technologies.
- **Domain-specific social, cooperative, and collaborative applications.** Including applications to healthcare, transportation, design, manufacturing, gaming, ICT4D, sustainability, education, accessibility, global collaboration, or other domains.
- **Ethics and policy implications.** Analysis of the implications of sociotechnical systems in social, cooperative and collaborative practices, as well as the algorithms that shape them.
- **CSCW and social computing systems based on emerging technologies.** Including mobile and ubiquitous computing, game engines, virtual worlds, multi-touch, novel display technologies, vision and gesture recognition, big data, MOOCs, crowd labor markets, SNSs, computer-aided or robotically-supported work, and sensing systems.
- **Crossing boundaries.** Studies, prototypes, or other investigations that explore interactions across fields of research, disciplines, distances, languages, generations, and cultures to help better understand how CSCW and social systems might help transcend social, temporal, and/or spatial boundaries.

Compare CHI, CSCW, ASSETS?

Conference Other Fields?

AAAI Conference on Artificial Intelligence

AAAI/ACM Conference on AI, Ethics, and Society (AIES)

American Medical Informatics Association (AMIA) Annual Symposium

CHI PLAY

ISLS

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Toward participatory research: "Nothing about us without us"

Challenges in this Field